

**Solutions of Part 2 Exam: Mathematics for
Microeconomics**

1. (20 marks) Consider the following quadratic form:

$$Q(x, y, z) = 5x^2 + 3y^2 + 4yz + 2xz$$

- (a) (4 marks) Identify the *symmetric matrix of coefficients* A associated with this quadratic form. That is

$$Q(x, y, z) = (x, y, z) A \begin{pmatrix} x \\ y \\ z \end{pmatrix}$$

Find all the leading submatrices.

Solutions: The matrix of coefficients is:

$$A = \begin{pmatrix} 5 & 0 & 1 \\ 0 & 3 & 2 \\ 1 & 2 & 0 \end{pmatrix}$$

It's leading sub-matrices are:

$$A_3 = \begin{pmatrix} 5 & 0 & 1 \\ 0 & 3 & 2 \\ 1 & 2 & 0 \end{pmatrix} \quad A_2 = \begin{pmatrix} 5 & 0 \\ 0 & 3 \end{pmatrix} \quad A_1 = (5)$$

- (b) (8 marks) Define what it means for the quadratic form $Q(x, y, z)$ to be *indefinite*. How would you be able to assess, in terms of the properties of A , whether $Q(x, y, z)$ is indefinite?

Solutions: A quadratic form is indefinite if $Q(x', y', z') < 0$ for some (x', y', z') and if $Q(x'', y'', z'') > 0$ for some (x'', y'', z'') . The quadratic form is indefinite when A is neither positive semi-definite (all leading minors are non-negative) nor negative semi-definite (all odd leading minors are non-positive and all even leading minors are non-negative).

- (c) (8 marks) Is the quadratic form $Q(x, y, z)$ *positive definite*, *negative definite*, *positive semi-definite*, *negative semi-definite* or *indefinite*?

Solutions: $Q(x, y, z)$ is indefinite since A is indefinite since the third and the first leading minors have opposite signs:

$$|A_1| = 5 > 0, \quad |A_2| = \begin{vmatrix} 5 & 0 \\ 0 & 3 \end{vmatrix} = 15 > 0, \quad |A| = \begin{vmatrix} 5 & 0 & 1 \\ 0 & 3 & 2 \\ 1 & 2 & 0 \end{vmatrix} = -23 < 0$$

2. (40 marks) Consider the function defined for $x > a$:

$$f(x) = -(\ln(x - a))^2$$

where $\ln(x)$ denotes the natural logarithm of the number x .

- (a) (8 marks) Find the *critical points* of $f(x)$ if any exists for $x > a$.

Solutions: The critical point of function $f(x) = -(\ln(x - a))^2$ is $x^* = a + 1$ and it is the solution to

$$\frac{df(x)}{dx} = -2 \frac{\ln(x - a)}{x - a} = 0 \quad (1)$$

- (b) (8 marks) Are these critical points *local maximizers*, *local minimizers* or *neither*?

Solutions: The critical point $x^* = a + 1$ of the function $f(x) = -(\ln(x - a))^2$ is a local maximizer since the sufficient second order condition for a local minimizer is satisfied at such point:

$$\frac{d^2f(x^*)}{dx^2} = 2 \frac{(\ln(x^* - a) - 1)}{(x^* - a)^2} = -2 < 0 \quad (2)$$

- (c) (8 marks) Are these critical points *global maximizers*, *global minimizers* or *neither*?

Solutions: The critical point $x^* = a + 1$ of the function $f(x) = -(\ln(x - a))^2$ is also a global maximizer since $f(x) \leq 0$ for any $x > a$ and since $f(x^*) = 0$.

- (d) (8 marks) Is $f(x)$ *concave*, *convex*, *quasi-concave*, *quasi-convex* or *neither of the above* for $x > a$?

Solutions: The concavity of function $f(x) = -(\ln(x - a))^2$ depends on the sign of $\ln(x - a) - 1$ by 2. Thus, the function is convex if $x \geq a + e$ and concave if $x \leq a + e$. Thus it is neither globally convex, nor globally concave. Moreover, since the first derivative $f'(x)$ is positive for any $x < x^*$ and negative for any $x > x^*$, the function is single peaked and thus it is quasi-concave, but not quasi-convex.

- (e) (8 marks) Let $x^*(a)$ denote the *global maximizer* of $f(x)$ for $x > a$. Use the envelope theorem to show how *maximum value function* responds to a marginal change in a . Also show how the *global maximizer* $x^*(a)$ is affected by marginal changes in a .

Solutions: By the envelope theorem we know that:

$$\frac{d}{da}f(x^*(a), a) = \frac{\partial}{\partial a}f(x^*(a), a) = 2\frac{\ln(x^*(a) - a)}{x^*(a) - a} = 0$$

Furthermore since $x^*(a) = a + 1$:

$$\frac{\partial}{\partial a}x^*(a) = 1$$

3. (40 marks) For $I > 0$, consider the problem:

$$\begin{aligned} \max_{x,y} \quad & F(x, y) = x^{1/3}y^{2/3} \\ \text{s.t.} \quad & 3x + y \leq I \\ & x \geq 0 \\ & y \geq 0 \end{aligned} \tag{3}$$

- (a) (8 marks) Establish whether the objective function and the constraints are respectively concave, convex, or neither. Which conditions on the concavity/convexity of such functions would suffice to guarantee that the Lagrangian is concave?

Solutions: The objective function is concave since it's Hessian is negative semi-definite for any $x, y \geq 0$:

$$d^2F(x, y) = \begin{pmatrix} -(2/9)x^{-5/3}y^{2/3} & (2/9)x^{-2/3}y^{-1/3} \\ (2/9)x^{-2/3}y^{-1/3} & -(2/9)x^{1/3}y^{-4/3} \end{pmatrix}$$

since $|d^2F(x, y)| = 0$, $-(2/9)x^{-5/3}y^{2/3} \leq 0$, and $-(2/9)x^{1/3}y^{-4/3} \leq 0$

The constraints are obviously linear (both convex and concave). If the objective function is concave and the constraints are convex, the Lagrangian will be concave, since it is a sum of concave functions.

- (b) (8 marks) Set up the constrained optimization problem using the Lagrangian function. Can you use the Lagrange method to solve this problem?

Solutions: The Lagrangian of the problem is

$$\mathcal{L}(x, y, \lambda_1, \lambda_2, \lambda_3) = x^{1/3}y^{2/3} - \lambda_1(3x + y - I) + \lambda_2 x + \lambda_3 y$$

Notice first that a simple contradiction argument proves that if $I > 0$ the optimal solution cannot have $x = 0$ and/or $y = 0$, as we could increase the value of the program by setting $x = y = I/4$.

The Lagrangian of the problem then becomes:

$$\mathcal{L}(x, y, \lambda_1, \lambda_2, \lambda_3) = x^{1/3}y^{2/3} - \lambda_1(3x + y - I)$$

The constraint set has a non-empty interior thus we can adopt the Kuhn-Tucker approach to solve the problem. Moreover, the gradient of the constraint $g(x, y) = 3x + y - I$ is:

$$\nabla g(x, y) = (3, 1)$$

therefore constraint qualifications for the Lagrange method are satisfied for every pair (x, y) such that $3x + y \leq I$. In other words, we can use the Lagrangian method to solve the constrained maximization problem.

- (c) (8 marks) Write down the necessary first order conditions for the solution of the constrained optimization problem.

Solutions: The necessary first order conditions for the solution of the constrained maximization problem are then:

$$\begin{aligned} 0 &= (1/3)(y/x)^{2/3} - 3\lambda_1 \\ 0 &= (2/3)(x/y)^{1/3} - \lambda_1 \\ 0 &= \lambda_1(3x + y - I) \\ 0 &\geq 3x + y - I \\ 0 &\leq \lambda_1 \end{aligned}$$

- (d) (4 marks) Are these conditions *necessary* or both *necessary and sufficient*?

Solutions: The necessary first order conditions for the solution of the constrained maximization problem are both necessary and sufficient for a maximum since the objective function is concave and the constraint function $g(x, y)$ is convex.

- (e) (4 marks) Solve the constrained optimization problem in terms of I .

Solutions: The solution to the constraint optimization problem is the solution to the necessary and sufficient conditions above and is $\lambda_1 = (4/81)^{1/3}$ and $(x^*(I), y^*(I)) = (I/9, 2I/3)$.

- (f) (8 marks) Let $x^*(I)$ and $y^*(I)$ be the solution of this problem. Find the slope of such functions with respect to I . Moreover, find how the maximum value function is affected by a change in I .

Solutions: Since the bordered Hessian is negative definite at the global optimum, the implicit function theorem applies. Thus, it is immediate to observe that:

$$\frac{\partial}{\partial I}x^*(I) = \frac{1}{9} \text{ and } \frac{\partial}{\partial I}y^*(I) = \frac{2}{3}.$$

Finally, observe that by the Envelope theorem:

$$\begin{aligned}\frac{d}{dI}F(x^*(I), y^*(I)) &= \frac{\partial}{\partial x}F(x^*(I), y^*(I))\frac{\partial}{\partial I}x^*(I) + \frac{\partial}{\partial y}F(x^*(I), y^*(I))\frac{\partial}{\partial I}y^*(I) = \\ &= \lambda_1 \left[\frac{\partial}{\partial x}g(x^*(I), y^*(I), I)\frac{\partial}{\partial I}x^*(I) + \frac{\partial}{\partial y}g(x^*(I), y^*(I), I)\frac{\partial}{\partial I}y^*(I) \right] = \\ &= \lambda_1 = (4/81)^{1/3}\end{aligned}$$

for $g(x, y, I) = 3x + y - I$.